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Statistical Models
Week 13: Model comparison and model selection

In every statistical analysis, we should critically assess our model choice. This week deals with model selection and validation tools for Bayesian analysis: Cross validation, Bayes factors, Stochastic Search Variable Selection, Information criteria DIC and WAIC, and posterior validation (goodness-of-fit).

Relevant Readings

- ☐ Hypothesis testing and Bayes factors: Reich & Ghosh, Section 5.2
- ☐ Cross validation: Reich & Ghosh, Section 5.1
- ☐ Model selection: Reich & Ghosh, Section 5.3, 5.5
- ☐ Goodness-of-fit: Reich & Ghosh, Section 5.6

Lecture Material

- Link to onenotes on Google drive
- Example for crossvalidation

Exercises

Problem 13.1: Air quality and cross validation

Inspect the `airquality` dataset in R (it is contained in base R). Read about its background.
We consider the following two models for ozone pollution:

$$\mathcal{M}_1: \quad \text{ozone}_i \sim \text{Normal}(\beta_1 + \beta_2 \text{solar.R}_i, \sigma^2)$$
$$\mathcal{M}_2: \quad \text{ozone}_i \sim \text{Normal}(\beta_1 + \beta_2 \text{solar.R}_i + \beta_3 \text{temp}_i + \beta_4 \text{wind}_i, \sigma^2)$$

a. Compare the two models using 5-fold cross validation. Specify and justify the priors you select for both models.

b. Fit model $\mathcal{M}_2$ to the data, and use posterior predictive checks to verify that the model fits well. If you find model misspecification, suggest (but do not fit) alternatives.

Remark: Exclude rows with missing data from this data frame before analysis - you can conveniently achieve this with function `na.omit`.
see Reich & Ghosh, Chapter 5, Exercises 1 and 2 (there is a typo in model $\mathcal{M}_2$)

Problem 13.2: Bayes factors and information criteria

Reich & Ghosh, Chapter 5, Exercise 4.
Use the "Mr. October" data (described in Section 2.4, Reich and Ghosh) $Y_1 = 563$, $N_1 = 2820$, $Y_2 = 10$, and $N_2 = 27$.

Compare the two models:

$$\mathcal{M}_1 : Y_1 \mid \lambda_1 \sim \text{Poisson}(N_1\lambda_1) \text{ and } Y_2 \mid \lambda_2 \sim \text{Poisson}(N_2\lambda_2),$$
$$\mathcal{M}_2 : Y_1 \mid \lambda_0 \sim \text{Poisson}(N_1\lambda_0) \text{ and } Y_2 \mid \lambda_0 \sim \text{Poisson}(N_2\lambda_0),$$

using Bayes factors, *DIC* and *WAIC*. Assume the Uniform$(0, c)$ prior for all λ_j and compare the results for $c = 1$ and $c = 10$.

Hints

- The "Mr October" data are explained in R&G, Chapter 2, Exercise 2.
- You will realize that there is not much difference in the results for $c = 1$ and $c = 10$. You can try with smaller values of c - what would that mean? Does it make sense?
- I have uploaded some code on Google drive (week 13). If you use it, you save a lot of time. Please spend that time on thinking more about this example:
  - why is the answer so stable when you use $c = 1$, $c = 10$, or even $c = 100$?
  - can you describe the posterior distribution of the parameter theoretically, without using MCMC?

Problem 13.3: Selecting a model for house prices

Reich & Ghosh, Chapter 5, Exercise 9.

This exercise extends Problem 12.4.

Download the Boston Housing Data in R from the Boston dataset (you can do this by `library(MASS); data(Boston)`). The response is `medv`, the median value of owner-occupied homes, and the other 13 variables are covariates that describe the neighborhood.

Use stochastic search variable selection (SSVS) to compute the most likely subset of the 13 covariates to include in the model and the marginal probability that each variable is included in the model. Clearly describe the model you fit including all prior distributions.

Problem 13.4: A model for NBA data, and posterior checks

cf Reich & Ghosh, Chapter 5, Exercises 7 + 8.

For the NBA free throw data described in Reich & Ghosh, Section 1.6, Exercise 17, assume that for player i, the number Y_i of clutch makes is distributed as $Y_i \mid p_i \sim \text{Binomial}(n_i, p_i)$, where n_i is the number of clutch attempts, and p_i is the clutch make probability. Compute the posterior probabilities of the models:

$$\mathcal{M}_1 : \text{logit}(p_i) = \beta_1 + \text{logit}(q_i)$$
$$\mathcal{M}_2 : \text{logit}(p_i) = \beta_1 + \beta_2 \text{logit}(q_i),$$

where q_i is the overall free throw percentage of player i.

a. Specify the priors you use for the models' parameters and discuss whether the results are sensitive to the prior.

b. Fit model $\mathcal{M}_2$ to the NBA data and use posterior predictive checks to verify the model fits well. If you find model misspecification, suggest (but do not fit) alternatives.

You can find the data as file `NBA-clutches.csv` in the data folder on Google Drive or download it here

Problem 13.5: MAP in linear models and their frequentist counterparts

Consider a linear normal model

$$Y_i \sim \text{Normal}(\sum_{j=1}^{p} x_{ij}\beta_j, \sigma^2)$$

where the Y_i, $i = 1, \dots n$ are independent, as usual. We assume independent priors for the coefficients $\beta_1, \dots, \beta_p$ and for σ, and for sake of simplicity, that the prior is the same for $\beta_1, \beta_2, \dots$ and β_p.
Show that the maximum a posteriori estimate for $\beta = (\beta_1, \dots, \beta_p)$ minimizes the following (penalized) residual sums of squares, depending on the prior:

$$\hat{\beta}_{\text{MAP}} = \text{argmin}_\beta \begin{cases} \|Y - X\beta\|^2, & \pi(\beta) = \text{Jeffreys' prior}, \\ \|Y - X\beta\|^2 - \lambda \sum_{j=1}^{p} \beta_j^2, & \pi(\beta) = \text{Gaussian prior}, \\ \|Y - X\beta\|^2 - \lambda \sum_{j=1}^{p} |\beta_j|, & \pi(\beta) = \text{double exponential prior}. \end{cases}$$

For the Gaussian prior, let $\beta_j \sim \text{Normal}(0, \tau^2)$, and for the double exponential prior, let $\pi(\beta_j) = \frac{\gamma}{2}\exp(-\gamma|\beta_j|)$.
In these two cases, express λ in terms of the prior's hyperparameters (τ^2 and γ, respectively).